Example 9.18

Construct the cost of living index number for 2011 on the basis of 2007 from the given data using family budget method.

Commodities	Price		Weights
	2007	2011	
A	350	400	40
B	175	250	35
C	100	115	15
D	75	105	20
E	60	80	25

Solution:

Commo-dities	Price		Weights (V)	$P=\frac{P_1}{P_0}\times 100$	PV
	2007 (P_0)	2011 (P_1)			
A	350	400	40	114.286	4571.44
B	175	250	35	142.857	4999.995
C	100	115	15	115	1725
D	75	105	20	140	2800
E	60	80	25	133.333	3333.325
		Total	135		17429.76

Table 9.17

Cost of Living Index Number

$$= \frac{\sum PV}{\sum V} = \frac{17429.76}{135} = 129.1093$$

Hence, the Cost of Living Index Number for a particular class of people for the year 2011 is increased by 29.1093% as compared to the year 2007.

Exercise 9.2

1. Define Index Number.
2. State the uses of Index Number.
3. Mention the classification of Index Number.
4. Define Laspeyre's price index number.
5. Explain Paasche's price index number.
6. Write note on Fisher's price index number.
7. State the test of adequacy of index number.
8. Define Time Reversal Test.
9. Explain Factor Reversal Test.
10. Define true value ratio.
11. Discuss about Cost of Living Index Number.
12. Define Family Budget Method.
13. State the uses of Cost of Living Index Number.
14. Calculate by a suitable method, the index number of price from the following data:

Commo-dity	2002		2012	
	Price	Quantity	Price	Quantity
A	10	20	16	10
B	12	34	18	42
C	15	30	20	26

15. Calculate price index number for 2005 by (a) Laspeyre's (b) Paasche's method

Commo-dity	1995		2005	
	Price	Quantity	Price	Quantity
A	5	60	15	70
B	4	20	8	35
C	3	15	6	20

16. Compute (i) Laspeyre's (ii) Paasche's (iii) Fisher's Index numbers for the 2010 from the following data.

Commo-dity	Price		Quantity	
	2000	2010	2000	2010
A	12	14	18	16
B	15	16	20	15
C	14	15	24	20
D	12	12	29	23

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